

Verification of Robustness Transfer Between Neural Networks

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Abstract. Neural networks have achieved remarkable success across a wide range of applications, but they remain vulnerable to *adversarial examples*—maliciously perturbed inputs designed to deceive the network. Verifying the robustness of neural networks against such attacks is a challenging task. In particular, verifying the *global robustness* of a network—ensuring robustness for any input under a given perturbation function and scale—is computationally difficult. In this work, we address the problem of *robustness transfer verification*. Given a source network N_1 with a known robustness guarantee and a target network N_2 , we compute a formal certificate guaranteeing N_2 's global robustness property. We encode this problem as a mixed-integer program (MIP). Our approach supports seven perturbation types and employs four complementary tightening techniques: adversarial warm-starting, perturbation-difference interval propagation, inter-network interval bounds, and dependency analysis. We provide a complete formal analysis, including a proof that the transfer MIP yields a sound lower bound on the robustness improvement, a counterexample demonstrating that this bound can be strict, and a modified formulation that provides a matching upper bound over a restricted domain. Finally, we present a method for computing a sound upper bound during inference.

Keywords: Neural Network Verification · Global Robustness · Constrained Optimization

1 Introduction

Deep neural networks are successful in various tasks but are also susceptible to adversarial examples: malicious input perturbations designed to deceive the network [10,11,9]. Many adversarial attacks target image classifiers and compute either an imperceptible change, bounded by a small ε with respect to an L_p norm, or a perceivable change obtained by perturbing a semantic feature of the input, such as brightness, translation, or occlusion.

A popular safety property for understanding a network's resilience to such attacks is *local robustness*. Local robustness is parameterized by an input and its neighborhood: for a network classifier, the goal is to prove that the network classifies all inputs in the neighborhood the same. Several local robustness verifiers have been introduced for checking robustness in an ε -ball [6,8,7]. However,

13 local robustness is limited to reasoning about a single neighborhood at a time.
 14 To reason about a network’s robustness over *all* possible inputs, VHAGaR [2]
 15 introduced the concept of *global robustness*: computing the *minimal globally ro-*
 16 *bust bound*, the smallest confidence above which the network is provably robust
 17 to a given perturbation.

18 In practice, neural networks are rarely deployed in isolation. A common sce-
 19 nario involves training a more accurate network N_2 from an existing network
 20 N_1 via fine-tuning, distillation, or retraining. A natural question arises: *does the*
 21 *robustness guarantee of N_1 transfer to N_2 ?* More precisely, if N_1 is known to
 22 be globally robust above some confidence threshold $\delta_1(N_1)$, can we certify N_2 ’s
 23 robustness without re-running the expensive global verification from scratch?

24 In this work, we address the problem of *robustness transfer verification*. Given
 25 a source network N_1 and a target network N_2 , we compute a confidence mar-
 26 gin δ_{diff} such that any input where N_2 ’s confidence exceeds N_1 ’s by more than
 27 δ_{diff} inherits N_1 ’s robustness guarantee. Intuitively, δ_{diff} captures the worst-case
 28 confidence gap between the two networks in the regime where N_1 is certifiably ro-
 29 bust but N_2 can still be fooled. This problem is challenging for two main reasons.
 30 First, it requires encoding three forward passes — $N_1(x)$, $N_2(x)$, and $N_2(x')$ —
 31 simultaneously in a single optimization problem. Second, the perturbation in-
 32 troduces complex dependencies between the clean and perturbed computations
 33 that must be captured precisely for the resulting bounds to be tight.

34 We extend VHAGaR with a *transfer mode* that encodes the robustness trans-
 35 fer problem as a mixed-integer program (MIP). Our MIP simultaneously models
 36 three forward passes — $N_1(x)$, $N_2(x)$, and $N_2(x')$ — linked by the perturba-
 37 tion constraint $x' \in \mathcal{I}_\varepsilon(x)$. To make the resulting MIP tractable, we introduce
 38 four complementary tightening techniques: (1) a *hyper-adversarial attack* that
 39 provides a warm-start solution via projected gradient descent, (2) *perturbation-*
 40 *difference interval propagation* that bounds the activation differences between
 41 clean and perturbed passes, (3) *inter-network interval bounds* exploiting the
 42 structural similarity between N_1 and N_2 , and (4) *dependency analysis* that iden-
 43 tifies monotone relationships between activations. Our approach currently sup-
 44 ports the L_∞ perturbation and six semantic feature perturbations.

45 We provide a complete formal analysis of the transfer MIP’s guarantees. We
 46 prove that the computed margin δ_{diff} provides a *sound lower bound*: $\delta_1(N_2) \geq$
 47 $\delta_1(N_1) + \delta_{\text{diff}}$, meaning N_2 ’s global robustness bound is at least the sum of N_1 ’s
 48 bound and the transfer margin. We then exhibit a concrete counterexample
 49 with two simple linear networks demonstrating that the reverse inequality is
 50 *false* in general — the bound is strict. Finally, we propose a modified transfer
 51 MIP, replacing the constraint that N_1 is robust with the constraint that N_1 is
 52 *foolable*, and prove that the resulting quantity $\hat{\delta}_{\text{diff}}$ satisfies the complementary
 53 upper bound $\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)$.

54 In summary, our main contributions are: (1) formalizing the robustness trans-
 55 fer problem and encoding it as a MIP with four tightening techniques (Section 6),
 56 (2) proving that the transfer MIP yields a sound lower bound on N_2 ’s robust-
 57 ness improvement over N_1 (Section 8), (3) exhibiting an explicit counterexample

showing the bound is strict, with a precise characterization of the gap (Section 8), and (4) designing a modified formulation that provides a matching upper bound over the joint-foolable domain (Section 8).

The rest of this paper is organized as follows. Section 2 provides background on network classifiers and formalizes the central quantities. Section 4 defines the supported perturbation models. Sections 5 and 6 present the standard-mode and transfer-mode MIP formulations, respectively. Section 7 discusses the two attack constraint variants. Section 8 provides the full formal analysis. Section 9 describes the four tightening techniques. Section 10 gives the complete MIP formulation, and Sections 11 and 12 present the runtime inference procedure and conclusions.

2 Preliminaries

In this section, we provide background on neural network classifiers and formalize the central quantities used throughout this paper. We begin with the definitions of networks and confidence margins, then introduce the notions of robustness and foolability that underlie both the standard-mode and transfer-mode formulations.

2.1 Neural Network Classifiers

A classifier maps an input into a score vector over a set of labels (also called classes) C , i.e., it is a function $N : [0, 1]^n \rightarrow \mathbb{R}^{|C|}$. Given an input x , its classification is the label with the maximal score: $c = \operatorname{argmax}(N(x))$. We focus on classifiers implemented as neural networks. A neural network consists of an input layer followed by L layers, where the last layer is the output layer. Layers consist of neurons, each connected to some or all neurons in the next layer.

We denote by $z_{m,k}$ the k -th neuron in layer m , for $m \in \{0, \dots, L\}$ (where 0 denotes the input layer) and $k \in \{1, \dots, k_m\}$. The input layer z_0 passes the input x to the next layer (i.e., $z_{0,k} = x_k$ for all $k \in [n]$). In a fully-connected layer, a neuron $z_{m,k}$ gets as input the outputs of all neurons in the preceding layer. It has a weight $w_{m,k,k'}$ for each input and a single bias $b_{m,k}$. Its function is the weighted sum of its inputs and bias followed by an activation function. Formally, the function of a non-input neuron is $z_{m,k} = \operatorname{ReLU}(\hat{z}_{m,k}) = \max(0, \hat{z}_{m,k})$, where $\hat{z}_{m,k}$ is the weighted sum: $\hat{z}_{m,k} = b_{m,k} + \sum_{k'=1}^{k_{m-1}} w_{m,k,k'} \cdot z_{m-1,k'}$. The network's parameters — the weights and biases — are determined by a training algorithm. The output layer consists of $|C|$ neurons, each outputting the score of one of the labels. We denote the output logits by $\hat{y}(x) = z^{[L]}(x)$.

Our definitions focus on fully-connected layers, but our implementation also supports convolutional layers and it is easily extensible to other layer types such as max-pooling layers [3].

96 2.2 Confidence Margin

97 The *confidence margin* captures how certain a network is in its classification.
 98 Intuitively, a high confidence margin means the network assigns the correct class
 99 a score well above all other classes, making it harder for a perturbation to change
 100 the classification.

101 **Definition 1 (Confidence margin).** *Given a network N , an input x , and a*
 102 *class $c \in C$, the class confidence of N in c at x is:*

$$\mathcal{C}(N, x, c) := \hat{y}_c(x) - \max_{k \neq c} \hat{y}_k(x).$$

103 *The confidence $\mathcal{C}(N, x, c) > 0$ if and only if N classifies x as c .*
 104 *Two-class setting. When there are only two output classes c_{tag} and c_{target} , we*
 105 *write $\mathcal{C}(N, x)$ as shorthand for $\mathcal{C}(N, x, c_{\text{tag}}) = \hat{y}_{c_{\text{tag}}}(x) - \hat{y}_{c_{\text{target}}}(x)$.*

106 2.3 Robustness and Foolability

107 We next define robustness and foolability with respect to a perturbation set.
 108 These notions form the building blocks of both the standard-mode and transfer-
 109 mode problems.

110 **Definition 2 ($(\varepsilon, c \rightarrow t)$ -robustness).** *Given a network N and a perturbation*
 111 *set $\mathcal{I}_\varepsilon(x)$ (defined per perturbation type in Section 4), an input x is $(\varepsilon, c \rightarrow t)$ -*
 112 *robust for N if $\forall x' \in \mathcal{I}_\varepsilon(x) : N(x') \neq t$. That is, no perturbation in the allowed*
 113 *set causes N to misclassify x as the target class t .*

114 **Definition 3 (N -foolability).** *An input $x \in \mathcal{X}$ is N -foolable if there exists a*
 115 *perturbation $\varepsilon \in \mathcal{E}$ with $x + \varepsilon \in \mathcal{X}$ such that $\mathcal{C}(N, x + \varepsilon) \leq -\eta$, where $\eta > 0$ is a*
 116 *small attack margin (typically 10^{-3}). We write $\mathcal{F}_N$ for the set of all N -foolable*
 117 *inputs.*

118 Intuitively, an input is foolable if the adversary can find a perturbation that
 119 causes the network to misclassify it. The set $\mathcal{F}_N$ contains all inputs that are
 120 vulnerable to adversarial attack under the given perturbation model.

121 2.4 MIP Encoding of ReLU

122 Our approach encodes the network’s computation as a mixed-integer program
 123 (MIP). The JuMP/Gurobi MIP [5,4] encodes each linear layer exactly and lin-
 124 earizes every ReLU via the standard big- M formulation with a binary variable
 125 $b \in \{0, 1\}$:

$$a = \max(0, z), \quad 0 \leq a \leq M b, \quad z \leq a, \quad a \leq z + M(1 - b).$$

126 This encoding assigns each non-input ReLU neuron a binary variable, whose
 127 domain is $\{0, 1\}$. Generally, the time complexity of a MIP problem over boolean
 128 and real variables is exponential in the number of boolean variables. To make the
 129 analysis more efficient, several works propose ways to identify boolean variables
 130 that can be removed (e.g., ReLUs whose inputs are non-positive or non-negative),
 131 thereby reducing the complexity [8].

2.5 Formal Definitions of the Central Quantities

We now formalize the three central MIP quantities. Throughout, we use: $\mathcal{X} = [0, 1]^n$ (input domain), $\mathcal{E} = [-p, p]^n$ (perturbation ball for radius $p > 0$), attack margin $\eta > 0$, and transition margin $\varepsilon_0 > 0$ (both typically 10^{-3}).

Definition 4 (Standard-mode value). *The standard-mode value of network N_i is:*

$$\delta_1(N_i) := \sup \{ \mathcal{C}(N_i, x) \mid x \in \mathcal{F}_i, \mathcal{C}(N_i, x) \geq 0 \}.$$

Intuitively, this is the maximum confidence N_i attains on any input that is correctly classified yet can still be misclassified by some perturbation $\varepsilon \in \mathcal{E}$. Any input with confidence above $\delta_1(N_i)$ is provably robust.

Definition 5 (Transfer MIP — original). *Given networks N_1 and N_2 with pre-computed $\delta_1(N_1)$, the transfer value is:*

$$\begin{aligned} \delta_{\text{diff}}^* := \sup \{ & \mathcal{C}(N_2, x) - \mathcal{C}(N_1, x) \mid \\ & \text{(C1) } \mathcal{C}(N_1, x) \geq \delta_1(N_1) + \varepsilon_0, \\ & \text{(C2) } \mathcal{C}(N_2, x) \geq \mathcal{C}(N_1, x), \\ & \text{(C3) } x \in \mathcal{F}_2 \}. \end{aligned} \quad (1)$$

Constraint C1 ensures x lies in the region where N_1 is certifiably robust. C2 ensures N_2 is at least as confident as N_1 . C3 ensures a perturbation exists that fools N_2 . The objective maximizes the confidence gap, finding the worst-case scenario for the transfer guarantee.

Definition 6 (Modified transfer MIP).

$$\begin{aligned} \hat{\delta}_{\text{diff}} := \sup \{ & \mathcal{C}(N_2, x) - \mathcal{C}(N_1, x) \mid \\ & (\widehat{\text{C1}}) \ x \in \mathcal{F}_1, \\ & \text{(C3) } x \in \mathcal{F}_2 \}. \end{aligned} \quad (2)$$

The key change: C1 is replaced by $\widehat{\text{C1}}$, which requires x to be N_1 -foolable rather than N_1 -robust.

Definition 7 (Joint-foolable set). $\mathcal{F}_{1 \wedge 2} := \mathcal{F}_1 \cap \mathcal{F}_2$: *the set of inputs simultaneously foolable by both networks.*

Definition 8 (Modified standard-mode value).

$$\hat{\delta}_1(N_2) := \sup \{ \mathcal{C}(N_2, x) \mid x \in \mathcal{F}_{1 \wedge 2}, \mathcal{C}(N_2, x) \geq 0 \}.$$

This is the maximum confidence N_2 achieves at inputs simultaneously foolable by both networks. By definition, $\hat{\delta}_1(N_2) \leq \delta_1(N_2)$, since restricting to the joint-foolable set can only decrease the supremum.

153 3 Global Robustness Bounds

154 In this section, we define our problem: certifying the *global robustness* of a net-
 155 work and transferring such a certificate from one network to another. We begin
 156 by lifting the per-input robustness of Definition 2 to a property of the entire net-
 157 work, the global robustness of a network above a confidence threshold. We then
 158 define the *minimal globally robust bound*, the smallest threshold above which the
 159 network is provably robust, and relate it to the standard-mode value of Defini-
 160 tion 4. Finally, we formalize the robustness transfer problem and explain why it
 161 is challenging.

162 **Global robustness.** Local robustness (Definition 2) reasons about a single
 163 input and its neighborhood. To reason about a network’s behavior over all inputs
 164 at once, we lift it to a property of the whole network. Intuitively, a network is
 165 globally robust above a confidence threshold τ if every input it classifies with
 166 confidence at least τ is robust to the perturbation. Formally:

167 **Definition 9 (Global robustness).** *Given a network N , a class pair (c, t) , a*
 168 *perturbation scale ε , and a threshold $\tau \geq 0$, the network N is $(\varepsilon, c \rightarrow t)$ -globally*
 169 *robust above τ if every input $x \in \mathcal{X}$ with $\mathcal{C}(N, x, c) \geq \tau$ is $(\varepsilon, c \rightarrow t)$ -robust:*

$$\forall x \in \mathcal{X} : \mathcal{C}(N, x, c) \geq \tau \implies \forall x' \in \mathcal{I}_\varepsilon(x) : N(x') \neq t.$$

170 A larger threshold τ certifies fewer inputs and therefore gives a weaker guar-
 171 antee, so the strongest meaningful guarantee is the smallest τ that still holds.
 172 For example, if N is $(\varepsilon, c \rightarrow t)$ -globally robust above $\tau = 0.5$, then no ε -bounded
 173 perturbation changes the classification of any input that N assigns to c with con-
 174 fidence at least 0.5. Inputs classified with confidence below τ carry no guarantee
 175 and may be vulnerable.

176 **The minimal globally robust bound.** A network that is globally robust
 177 above τ is also globally robust above any $\tau' \geq \tau$, so the property holds for
 178 a whole range of thresholds. The strongest such guarantee corresponds to the
 179 smallest threshold that still holds, which we call the minimal globally robust
 180 bound. Intuitively, it is the tightest confidence level above which the network
 181 is provably robust, so that every input classified with higher confidence is safe.
 182 Formally:

183 **Definition 10 (Minimal globally robust bound).** *The minimal globally*
 184 *robust bound of N for $(\varepsilon, c \rightarrow t)$, denoted $\delta_1(N)$, is the smallest threshold above*
 185 *which N is globally robust:*

$$\delta_1(N) := \inf\{\tau \geq 0 : N \text{ is } (\varepsilon, c \rightarrow t)\text{-globally robust above } \tau\}.$$

186 This threshold coincides with the standard-mode value of Definition 4: the
 187 largest confidence attained by an input that N classifies correctly yet some per-
 188 turbation can still fool. Any input whose confidence exceeds $\delta_1(N)$ is provably
 189 robust, while arbitrarily close to the bound there is an input that some per-
 190 turbation fools. Section 5 shows how to compute $\delta_1(N)$ as the optimum of a
 191 mixed-integer program.

Robustness transfer. In practice, a target network N_2 is often trained from a source network N_1 by fine-tuning or distillation, and we already know N_1 's minimal globally robust bound $\delta_1(N_1)$. Recomputing the bound for N_2 from scratch is expensive, so we instead ask whether N_1 's guarantee transfers to N_2 . The goal is to compute a confidence margin δ_{diff} that relates the two bounds, certifying N_2 from N_1 rather than verifying N_2 anew.

Definition 11 (Robustness transfer problem). *Given a source network N_1 with a known minimal globally robust bound $\delta_1(N_1)$ and a target network N_2 , the robustness transfer problem is to compute a margin δ_{diff} for which*

$$\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}.$$

Intuitively, any input where N_2 's confidence exceeds N_1 's by more than δ_{diff} inherits N_1 's robustness guarantee, and δ_{diff} captures the worst-case confidence gap between the two networks in the regime where N_1 is certifiably robust but N_2 can still be fooled. The transfer-mode formulation of Section 6 computes such a margin as the optimum δ_{diff}^* of a single mixed-integer program, and Section 8 proves that this optimum satisfies the bound. The margin is obtained without re-solving the standard-mode problem for N_2 , which is the main practical benefit of the transfer approach.

This problem is challenging for two reasons. First, it requires encoding three forward passes, $N_1(x)$, $N_2(x)$, and $N_2(x')$, simultaneously in a single optimization problem, where x' is a perturbation of x drawn from $\mathcal{I}_\varepsilon(x)$. Second, the perturbation introduces complex dependencies between the clean and perturbed computations that must be captured precisely for the resulting margin to be tight.

4 Perturbation Models

In this section, we define the perturbation models supported by our framework. A perturbation is a function $f_P(x, \mathcal{E})$, defining how an input x is perturbed given a perturbation amount specified by a vector $\mathcal{E}$. All perturbations are parameterized by a scalar $\varepsilon > 0$ bounding their magnitude. For each perturbation type, we also define the *perturbation-difference interval* $[I^{\text{dn}}, I^{\text{up}}] \in \mathbb{R}^n \times \mathbb{R}^n$, which captures the element-wise range of $x' - x$ at the input layer. These intervals are propagated layer-by-layer to tighten the MIP (Section 9.2).

4.1 ℓ_∞ Perturbation

The ℓ_∞ perturbation has a perturbation limit $\varepsilon \in [0, 1]$ and is defined by a series of perturbations $\varepsilon_i \in [-\varepsilon, \varepsilon]$, for $i \in [n]$. The perturbation set is:

$$\mathcal{I}_\varepsilon(x) = \{x' : \|x' - x\|_\infty \leq \varepsilon, x' \in [0, 1]^n\}.$$

In the MIP encoding, this translates to: $x_i - \varepsilon \leq x'_i \leq x_i + \varepsilon$ for every pixel i , with clipping to $[0, 1]$.

The perturbation-difference interval is: $I_i^{\text{dn}} = -\varepsilon$, $I_i^{\text{up}} = +\varepsilon$ for all i .

229 4.2 Brightness Perturbation

230 Brightness is defined by a brightness level $e \in [0, \varepsilon]$ such that $x'_i = x_i + e$ for all
 231 pixels i . The perturbation set is:

$$\mathcal{I}_\varepsilon(x) = \{x' : x' = x + e \mathbf{1}, e \in [0, \varepsilon]\}.$$

232 In the MIP encoding, we add a scalar variable $e \in [0, \varepsilon]$ and set $x'_i = x_i + e$.

233 Since $e \geq 0$, the brightness shift is non-negative. The perturbation-difference
 234 interval is therefore: $I_i^{\text{dn}} = 0$, $I_i^{\text{up}} = +\varepsilon$ for all i .

235 4.3 Contrast Perturbation

236 Contrast is defined by a scaling factor $e \in [1, 1 + \varepsilon]$ such that $x'_i = e \cdot x_i$. The
 237 perturbation set is:

$$\mathcal{I}_\varepsilon(x) = \{x' : x' = e \cdot x, e \in [1, 1 + \varepsilon]\}.$$

238 In the MIP encoding, we add a scalar variable $e \in [1, 1 + \varepsilon]$ and set $x'_i = e \cdot x_i$. Note
 239 that this introduces a bilinear constraint, which requires the Gurobi non-convex
 240 mode.

241 Since $e \geq 1$ and $x_i \geq 0$, we have $x'_i - x_i = (e - 1)x_i \in [0, \varepsilon]$. The perturbation-
 242 difference interval is an over-approximation: $I_i^{\text{dn}} = 0$, $I_i^{\text{up}} = +\varepsilon$ for all i .

243 4.4 Translation Perturbation

244 Translation is defined by shift coordinates (d_h, d_w) with $|d_h|, |d_w| \leq k$ pixels,
 245 and vacated positions are zero-padded. The shift direction is selected by binary
 246 variables $b_h, b_w \in \{0, 1\}$ and offset variables $d_h, d_w \in [0, k]$.

247 Pixel values lie in $[0, 1]$. After translation, the worst-case difference at any
 248 pixel is $x'_i - x_i \in [-x_i, 1 - x_i] \subseteq [-1, 1]$. We use the conservative perturbation-
 249 difference interval: $I_i^{\text{dn}} = -1$, $I_i^{\text{up}} = +1$ for all i .

250 4.5 Patch Perturbation

251 A rectangular patch $P \subseteq [n]$ of pixels is perturbed by $\pm\varepsilon$, while pixels outside
 252 P are unchanged:

$$x'_i = x_i + e_i \varepsilon, \quad e_i \in [-1, +1], \quad i \in P; \quad x'_i = x_i, \quad i \notin P.$$

253 The perturbation-difference interval reflects this structure:

$$I_i^{\text{dn}} = \begin{cases} -\varepsilon & i \in P \\ 0 & i \notin P, \end{cases} \quad I_i^{\text{up}} = \begin{cases} +\varepsilon & i \in P \\ 0 & i \notin P. \end{cases}$$

254 4.6 Occlusion Perturbation

255 An occlusion perturbation sets a rectangular patch P to zero, while pixels outside
256 P are unchanged:

$$x'_i = 0, \quad i \in P; \quad x'_i = x_i, \quad i \notin P.$$

257 Because $x_i \in [0, 1]$ and $x'_i = 0$ inside P , the difference $x'_i - x_i = -x_i \in [-1, 0]$.
258 Outside P , the difference is exactly zero. The perturbation-difference interval is
259 therefore:

$$I_i^{\text{dn}} = \begin{cases} -1 & i \in P \\ 0 & i \notin P, \end{cases} \quad I_i^{\text{up}} = 0, \quad \forall i.$$

260 4.7 Summary

261 Table 1 summarizes the perturbation sets and their corresponding input-layer
262 difference intervals.

Table 1. Summary of perturbation sets and input-layer difference intervals.

Perturbation	Perturbation set $\mathcal{I}_\varepsilon(x)$	I_i^{dn}	I_i^{up}
ℓ_∞	$\ x' - x\ _\infty \leq \varepsilon$	$-\varepsilon$	$+\varepsilon$
Brightness	$x' = x + e, e \in [0, \varepsilon]$	0	$+\varepsilon$
Contrast	$x' = e \cdot x, e \in [1, 1 + \varepsilon]$	0	$+\varepsilon$
Translation	pixel shift $\leq k$ px, zero-pad	-1	+1
Patch	$\pm\varepsilon$ inside patch P	$-\varepsilon \cdot \mathbf{1}_P$	$+\varepsilon \cdot \mathbf{1}_P$
Occlusion	zero inside patch P	$-\mathbf{1}_P$	0

263 5 Standard Mode: Computing the Global Robustness 264 Bound

265 In this section, we present the standard-mode formulation, which computes the
266 global robustness bound for a single network. This bound serves as the founda-
267 tion for the transfer-mode analysis in Section 6.

268 5.1 Problem Definition

269 We fix a source class c and a target class t . The standard-mode MIP maximizes
270 the confidence margin of the *clean* input x while requiring that a *perturbed* input
271 x' is misclassified as t :

$$\delta_1 = \max_{x, x'} \mathcal{C}(N_1, x, c) \quad \text{subject to} \quad x' \in \mathcal{I}_\varepsilon(x), \quad \hat{y}_t(x') \geq \hat{y}_k(x') \quad \forall k \neq t. \quad (3)$$

Intuitively, the solution δ_1 is the highest confidence at which the network can still be fooled. Any input classified with confidence above δ_1 is provably robust: no perturbation in $\mathcal{T}_\varepsilon(x)$ can cause misclassification as t . This is equivalent to Definition 4: $\delta_1 = \delta_1(N_1)$ in the formal notation.

5.2 MIP Encoding of the Confidence Margin

The confidence margin involves $\max_{k \neq c} \hat{y}_k$, which is non-linear. We linearize it by introducing one binary variable $b_k \in \{0, 1\}$ per class $k \neq c$, indicating whether class k achieves the maximum:

$$\begin{aligned} \delta &= \hat{y}_c - m, \\ m &\geq \hat{y}_k \quad \forall k \neq c, \\ m &\leq \hat{y}_k + M(1 - b_k) \quad \forall k \neq c, \\ \sum_{k \neq c} b_k &= 1, \end{aligned}$$

where M is a sufficiently large constant. The solver then maximizes δ . This encoding is exact: the binary variables ensure that m equals the maximum logit among all non- c classes, and the MIP objective finds the maximum confidence at which an adversarial perturbation exists.

6 Transfer Mode: Verifying Robustness Across Networks

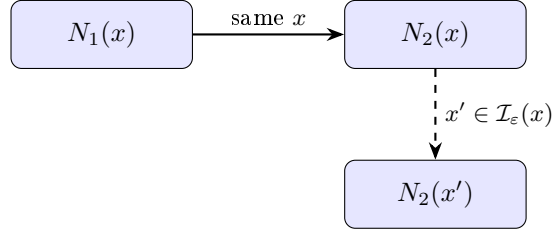
In this section, we present our main contribution: the transfer-mode formulation. We begin by motivating the problem, then describe our three-network MIP encoding, state the transfer MIP, and prove its correctness.

6.1 Motivation

Training a more accurate network N_2 from N_1 (e.g., via distillation or fine-tuning) often increases prediction confidence on correctly classified inputs. A natural question is: *by how much can N_2 's confidence exceed N_1 's in the worst case, in the regime where N_1 is already certifiably robust but N_2 can still be fooled?* The answer to this question determines whether N_1 's robustness guarantee transfers to N_2 .

6.2 Three-Network Encoding

To answer this question, the transfer mode encodes **three** forward passes simultaneously in a single MIP: $N_1(x)$, $N_2(x)$, and $N_2(x')$.



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296 Each forward pass uses a unique variable-name prefix to distinguish the three
 297 copies. The clean input x is shared between $N_1(x)$ and $N_2(x)$, and the perturbed
 298 input x' is linked to x via the perturbation constraint. Note that $N_1(x')$ is *not*
 299 encoded: the transfer MIP only requires N_1 's confidence on the clean input
 300 to establish the robustness baseline, and N_2 's computation on both the clean
 301 and perturbed inputs to verify whether the attack succeeds. This saves all of
 302 the ReLU binary variables that would be needed for a fourth network pass,
 303 significantly reducing the problem size.

304 6.3 Transfer MIP Formulation

305 Let $c_1 = \mathcal{C}(N_1, x, c)$, $c_2 = \mathcal{C}(N_2, x, c)$, and $c'_2 = \mathcal{C}(N_2, x', c)$. The transfer MIP is:

$$\delta_{\text{diff}} = \max_{x, x', c} (c_2 - c_1) \quad \text{s.t.} \quad \begin{cases} x' \in \mathcal{I}_\epsilon(x) & \text{(perturbation constraint)} \\ c_1 \geq \delta_1 + \epsilon_0 & \text{(T1: } N_1 \text{ is certifiably robust)} \\ c_2 - c_1 \geq 0 & \text{(T2: } N_2 \text{ at least as confident as } N_1) \\ \text{attack constraint on } N_2(x') & \text{(T3: } N_2 \text{ can be fooled)} \end{cases} \quad (4)$$

306 where $\epsilon_0 = 10^{-3}$ is a strict-inequality guard. In the formal notation of Sec-
 307 tion 2.5, this is exactly the transfer MIP of Definition 5, with $\delta_{\text{diff}} = \delta_{\text{diff}}^*$.

308 The constraint T1 restricts the search to inputs where N_1 is certifiably robust
 309 (its confidence exceeds the standard-mode bound δ_1). Constraint T2 focuses on
 310 the regime where N_2 is more confident than N_1 . Constraint T3 requires that a
 311 perturbation exists that fools N_2 , making the transfer problem non-trivial. The
 312 encoding of T3 depends on the chosen attack mode (Section 7).

313 6.4 Correctness of the Transfer Guarantee

314 We now prove that the transfer MIP provides a sound robustness certificate.
 315 The key insight is that any input violating the transfer guarantee would be a
 316 feasible solution to the MIP with objective value exceeding δ_{diff} , contradicting
 317 its optimality.

318 **Theorem 1 (Transfer Robustness).** *Let δ_1 be the standard-mode value for*
 319 *N_1 on perturbation $(\epsilon, c \rightarrow t)$. Let δ_{diff} be the optimal value of the transfer*
 320 *MIP (4). Then for any input x :*

$$\mathcal{C}(N_1, x, c) \geq \delta_1 \text{ and } \mathcal{C}(N_2, x, c) - \mathcal{C}(N_1, x, c) > \delta_{\text{diff}} \implies x \text{ is } (\epsilon, c \rightarrow t)\text{-robust for } N_2.$$

Proof. Suppose for contradiction that x is *not* $(\varepsilon, c \rightarrow t)$ -robust for N_2 . Then there exists $x' \in \mathcal{I}_\varepsilon(x)$ such that $N_2(x') = t$, meaning the attack constraint T3 holds.

Since $\mathcal{C}(N_1, x, c) \geq \delta_1$, constraint T1 is satisfied (with the ϵ_0 guard). Since $\mathcal{C}(N_2, x, c) - \mathcal{C}(N_1, x, c) > \delta_{\text{diff}} \geq 0$, constraint T2 is also satisfied.

Therefore (x, x') is a feasible solution to the transfer MIP (4) with objective value $c_2 - c_1 > \delta_{\text{diff}}$, contradicting the fact that δ_{diff} is the maximum feasible objective value.

Corollary 1 (Single-threshold inference). *Define $\Delta := \delta_1 + \delta_{\text{diff}}$. If $\mathcal{C}(N_1, x, c) \geq \delta_1$ and $\mathcal{C}(N_2, x, c) > \Delta$, then x is $(\varepsilon, c \rightarrow t)$ -robust for N_2 .*

This corollary shows that at inference time, the transfer guarantee reduces to checking two confidence thresholds, requiring only two forward passes and no additional optimization.

Remark 1. Checking $\mathcal{C}(N_2, x, c) > \Delta$ alone (without verifying N_1 's threshold) is *unsound*: constraint T1 would not be guaranteed, making the transfer MIP potentially infeasible and δ_{diff} meaningless. Both conditions must be verified.

6.5 Relationship to the Formal Quantities

The transfer MIP (4) computes $\delta_{\text{diff}} = \delta_{\text{diff}}^*$, which satisfies $\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*$ (Theorem 2 in Section 8). This is a *lower bound*: knowing δ_{diff} certifies that N_2 's standard-mode value is at least $\delta_1 + \delta_{\text{diff}}$, but N_2 's actual standard-mode value may be strictly larger. The precise gap and a counterexample are analyzed in Section 8.

7 Attack Constraint Variants

In this section, we describe the two variants for encoding constraint T3 (“ N_2 is fooled by x ”) in the transfer MIP. The choice of variant determines whether the adversary targets a specific class or any misclassification.

7.1 Targeted Attack

In the targeted variant, the adversary forces N_2 to predict a specific target class t . Formally, the constraint requires:

$$\hat{y}_t(N_2(x')) \geq \hat{y}_k(N_2(x')) \quad \forall k \neq t, \quad (5)$$

equivalently $\mathcal{C}(N_2, x', t) \geq 0$. This is the more restrictive variant: the adversary must not only cause a misclassification but must steer the network toward a particular wrong class.

353 7.2 Untargeted Attack

354 In the untargeted variant, the adversary need only reduce N_2 's confidence in the
 355 true class c to negative, causing any misclassification:

$$\mathcal{C}(N_2, x', c) \leq 0. \quad (6)$$

356 7.3 Comparison

357 Table 2 compares the two variants. The targeted variant has a smaller feasible
 358 region (the adversary must achieve a specific misclassification), leading to a larger
 359 or equal δ_{diff} and thus a stronger transfer guarantee. The untargeted variant has
 360 a larger feasible region, yielding a smaller or equal δ_{diff} and a weaker but more
 361 general guarantee.

Table 2. Comparison of targeted and untargeted attack constraints.

Property	Targeted	Untargeted
Attack type	$c \rightarrow t$ (specific class)	$c \rightarrow \text{any}$
Constraint on $N_2(x')$	$\hat{y}_t \geq \hat{y}_k, \forall k \neq t$	$\hat{y}_c < \hat{y}_k$ for some k
Feasible region	Smaller (harder to satisfy)	Larger (easier)
Resulting δ_{diff}	Larger or equal	Smaller or equal
Inference strength	Stronger guarantee	Weaker guarantee
Typical use	Specific adversary model	General robustness

362 8 Formal Analysis

363 In this section, we provide a rigorous analysis of the relationship between the
 364 standard-mode values $\delta_1(N_1)$, $\delta_1(N_2)$, and the transfer quantity δ_{diff}^* . We prove
 365 that the transfer MIP provides a sound lower bound, exhibit a counterexample
 366 showing the bound is strict, analyze the root cause of the gap, and propose a
 367 modified formulation with a matching upper bound. Throughout this section,
 368 we use the two-class confidence $\mathcal{C}(N, x) = \hat{y}_{c_{\text{tag}}}(x) - \hat{y}_{c_{\text{target}}}(x)$ for compactness.

369 8.1 The Transfer MIP Provides a Lower Bound

370 We first show that the transfer value δ_{diff}^* provides a sound lower bound on the
 371 improvement in robustness from N_1 to N_2 . Intuitively, any feasible point of the
 372 transfer MIP is also a feasible point of the standard-mode problem for N_2 , so
 373 the transfer MIP explores a subset of the relevant search space.

374 **Theorem 2 (Lower bound on $\delta_1(N_2)$).** *If the transfer MIP (1) is feasible,*
 375 *then*

$$\boxed{\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*}$$

Proof. Fix $\epsilon > 0$ and let x^* be a nearly-optimal feasible point of (1), with ε^* the corresponding attack perturbation (from C3). By feasibility:

$$\mathcal{C}(N_1, x^*) \geq \delta_1(N_1) + \varepsilon_0, \quad (\text{C1})$$

$$\mathcal{C}(N_2, x^*) \geq \mathcal{C}(N_1, x^*), \quad (\text{C2})$$

$$x^* \in \mathcal{F}_2 \text{ via } \varepsilon^*, \quad (\text{C3})$$

$$\mathcal{C}(N_2, x^*) - \mathcal{C}(N_1, x^*) \geq \delta_{\text{diff}}^* - \epsilon. \quad (\text{obj})$$

Step 1. We lower-bound $\mathcal{C}(N_2, x^*)$:

$$\begin{aligned} \mathcal{C}(N_2, x^*) &= \mathcal{C}(N_1, x^*) + (\mathcal{C}(N_2, x^*) - \mathcal{C}(N_1, x^*)) \\ &\geq (\delta_1(N_1) + \varepsilon_0) + (\delta_{\text{diff}}^* - \epsilon) \geq \delta_1(N_1) + \delta_{\text{diff}}^* - \epsilon. \end{aligned}$$

Step 2. We show x^* is feasible for the standard-mode MIP on N_2 . By C2 and C1, $\mathcal{C}(N_2, x^*) \geq \mathcal{C}(N_1, x^*) \geq \delta_1(N_1) + \varepsilon_0 > 0$, so N_2 correctly classifies x^* . Combined with C3 ($x^* \in \mathcal{F}_2$), the pair (x^*, ε^*) is feasible for the standard-mode MIP on N_2 . Hence:

$$\delta_1(N_2) \geq \mathcal{C}(N_2, x^*) \geq \delta_1(N_1) + \delta_{\text{diff}}^* - \epsilon.$$

Since $\epsilon > 0$ is arbitrary, we conclude $\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*$.

Remark 2 (The gap is always positive). Let x^* achieve the supremum in (1). The proof shows:

$$\delta_1(N_2) - (\delta_1(N_1) + \delta_{\text{diff}}^*) \geq \mathcal{C}(N_1, x^*) - \delta_1(N_1) \geq \varepsilon_0 > 0.$$

The gap is always positive because constraint C1 forces $\mathcal{C}(N_1, x^*) > \delta_1(N_1)$. This surplus $\mathcal{C}(N_1, x^*) - \delta_1(N_1)$ is the precise “slack” that makes the theorem strict.

8.2 The Reverse Inequality Fails: A Counterexample

We next show that the reverse inequality, $\delta_1(N_1) + \delta_{\text{diff}}^* \geq \delta_1(N_2)$, is false in general. We exhibit two explicit linear networks that produce a strict gap.

Network Definitions We consider two linear networks over $\mathcal{X} = [0, 1]^2$, with perturbation radius $p = 0.1$, and margins $\eta = \varepsilon_0 = 10^{-3}$.

	Network Logits	Confidence margin $\mathcal{C}(N, \cdot)$
N_1	$[x_1, x_2]$	$x_1 - x_2$
N_2	$[2x_1, 5x_2]$	$2x_1 - 5x_2$

Both are two-layer networks: an identity first layer, an inactive ReLU (since $x_1, x_2 \geq 0$), and a diagonal second layer (diag(1, 1) for N_1 , diag(2, 5) for N_2). Despite their simplicity, these networks suffice to demonstrate that the gap can be substantial.

Foolability Conditions We derive the foolability conditions for each network.
 N_1 is fooled at $x + \varepsilon$ if and only if $(x_2 + \varepsilon_2) - (x_1 + \varepsilon_1) \geq \eta$, i.e., $\varepsilon_2 - \varepsilon_1 \geq (x_1 - x_2) + \eta$. The maximum of $\varepsilon_2 - \varepsilon_1$ over $\mathcal{E}$ is $2p = 0.2$, so:

$$x \in \mathcal{F}_1 \iff x_1 - x_2 \leq 2p - \eta = 0.199.$$

N_2 is fooled at $x + \varepsilon$ if and only if $5(x_2 + \varepsilon_2) - 2(x_1 + \varepsilon_1) \geq \eta$, i.e., $5\varepsilon_2 - 2\varepsilon_1 \geq (2x_1 - 5x_2) + \eta$. The maximum of $5\varepsilon_2 - 2\varepsilon_1$ over $\mathcal{E}$ is $5p + 2p = 0.7$, so:

$$x \in \mathcal{F}_2 \iff 2x_1 - 5x_2 \leq 5p + 2p - \eta = 0.699.$$

Computing the Quantities We now compute $\delta_1(N_1)$, $\delta_1(N_2)$, and δ_{diff}^* for our counterexample networks.

Standard-mode value $\delta_1(N_1)$. We maximize $x_1 - x_2$ subject to $x_1 - x_2 \leq 0.199$ and $x \in [0, 1]^2$: $\delta_1(N_1) = 0.199$, achieved at $x^{(1)} = (1, 0.801)$.

Standard-mode value $\delta_1(N_2)$. We maximize $2x_1 - 5x_2$ subject to $2x_1 - 5x_2 \leq 0.699$ and $x \in [0, 1]^2$: $\delta_1(N_2) = 0.699$, achieved at $x^{(2)} = (1, 0.2602)$ with attack $\varepsilon^* = (-0.1, +0.1)$.

Transfer value δ_{diff}^ .* The objective is $\mathcal{C}(N_2, x) - \mathcal{C}(N_1, x) = (2x_1 - 5x_2) - (x_1 - x_2) = x_1 - 4x_2$. The constraints are:

$$\text{C1: } x_1 - x_2 \geq 0.200,$$

$$\text{C2: } x_1 - 4x_2 \geq 0,$$

$$\text{C3: } 2x_1 - 5x_2 \leq 0.699.$$

Setting $x_2 = 0$ (which minimizes the $-4x_2$ penalty), C1 gives $x_1 \geq 0.200$ and C3 gives $x_1 \leq 0.3495$. Maximizing x_1 yields:

$$x^* = (0.3495, 0), \quad \delta_{\text{diff}}^* = x_1^* - 4 \cdot 0 = \mathbf{0.3495}.$$

Note that $\mathcal{C}(N_1, x^*) = 0.3495$, which is substantially larger than $\delta_1(N_1) = 0.199$. This is the source of the gap.

The Counterexample

Quantity	Value
$\delta_1(N_1)$	0.199
δ_{diff}^*	0.3495
$\delta_1(N_1) + \delta_{\text{diff}}^*$	0.5485
$\delta_1(N_2)$	0.699
Gap: $\delta_1(N_2) - (\delta_1(N_1) + \delta_{\text{diff}}^*)$	0.1505

$$\delta_1(N_1) + \delta_{\text{diff}}^* = 0.5485 < 0.699 = \delta_1(N_2).$$

The gap equals exactly $\mathcal{C}(N_1, x^*) - \delta_1(N_1) = 0.3495 - 0.199 = 0.1505$, consistent with Remark 2.

415 8.3 Root-Cause Analysis

416 Theorem 2 and the counterexample together establish that $\delta_1(N_1) + \delta_{\text{diff}}^*$ is a
 417 *strict lower bound* on $\delta_1(N_2)$, with gap:

$$\text{gap} = \underbrace{\mathcal{C}(N_1, x^*)}_{\geq \delta_1(N_1) + \varepsilon_0 \text{ (C1)}} - \delta_1(N_1) \geq \varepsilon_0 > 0.$$

418 The structural reason is the following. The transfer MIP searches for an
 419 x^* that simultaneously satisfies C1 (high N_1 -confidence) and C3 (N_2 -foolable).
 420 Because N_1 and N_2 have *different* decision boundaries, the N_2 -foolable region
 421 can contain points with N_1 -confidence far above $\delta_1(N_1)$. At such an x^* , the
 422 quantity $\mathcal{C}(N_2, x^*) = \mathcal{C}(N_1, x^*) + \delta_{\text{diff}}^*$ is large, but the formula $\delta_1(N_1) + \delta_{\text{diff}}^*$
 423 loses the surplus $\mathcal{C}(N_1, x^*) - \delta_1(N_1)$.

424 Equivalently, the transfer MIP’s feasible set $\{x : \text{C1+C3}\}$ is a *proper subset*
 425 of the standard-mode feasible set for N_2 ($\{x \in \mathcal{F}_2\}$):

$$\delta_1(N_2) = \sup_{x \in \mathcal{F}_2} \mathcal{C}(N_2, x) \geq \sup_{x : \text{C1+C3}} \mathcal{C}(N_2, x) \geq \sup_{x : \text{C1+C2+C3}} [\mathcal{C}(N_1, x) + \delta_{\text{diff}}^*] \geq \delta_1(N_1) + \delta_{\text{diff}}^*.$$

426 The first inequality can be strict because $\mathcal{F}_2$ contains points with high $\mathcal{C}(N_2, \cdot)$
 427 that *violate* C1 (i.e., they are not N_1 -robust).

428 8.4 A Modified Formulation with a Matching Upper Bound

429 **Motivation** Remark 2 reveals that the gap is driven by C1 forcing $\mathcal{C}(N_1, x^*) >$
 430 $\delta_1(N_1)$. To close the gap, we replace C1 with its logical opposite: instead of
 431 requiring N_1 to be *robust* at x , we require N_1 to be *foolable* at x . This simulta-
 432 neously restricts the standard-mode problem for N_2 to the same joint-foolability
 433 region.

434 The modified transfer MIP is Definition 6, with modified standard-mode
 435 value $\hat{\delta}_1(N_2)$ (Definition 8). Note that $\hat{\delta}_1(N_2) \leq \delta_1(N_2)$ by definition: restricting
 436 to $\mathcal{F}_{1 \wedge 2} \subseteq \mathcal{F}_2$ can only decrease the supremum. The gap $\delta_1(N_2) - \hat{\delta}_1(N_2)$ mea-
 437 sures the advantage N_2 has at points where it is foolable but N_1 is not — the
 438 “exclusive vulnerability” of N_2 .

439 **MIP Encoding** The constraint $\widehat{\text{C1}}$ introduces a second perturbation variable
 440 $\varepsilon^{(1)} \in \mathcal{E}$ with the foolability constraint $\mathcal{C}(N_1, x + \varepsilon^{(1)}) \leq -\eta$, encoded exactly
 441 as in the standard-mode MIP for N_1 . Constraint C3 is encoded as before. The
 442 resulting program has $O(|N_1| + |N_2|)$ binary variables and is solvable by standard
 443 MIP solvers.

444 Main Result

445 **Theorem 3 (Upper bound on the joint standard-mode value).**

$$\boxed{\delta_1(N_1) + \delta_{\text{diff}} \geq \hat{\delta}_1(N_2).}$$

446 *Proof.* Fix $\epsilon > 0$ and let x^{**} be a nearly-optimal feasible point of $\hat{\delta}_1(N_2)$ (Defi-
 447 nition 8), i.e., $x^{**} \in \mathcal{F}_{1 \wedge 2}$, $\mathcal{C}(N_2, x^{**}) \geq 0$, $\mathcal{C}(N_2, x^{**}) \geq \hat{\delta}_1(N_2) - \epsilon$.

448 **Step 1.** Since $x^{**} \in \mathcal{F}_1$, by Definition 4, $\mathcal{C}(N_1, x^{**}) \leq \delta_1(N_1)$.

449 **Step 2.** Since $x^{**} \in \mathcal{F}_1$ satisfies $\widehat{C1}$ and $x^{**} \in \mathcal{F}_2$ satisfies C3, x^{**} is feasible
 450 for (2). Hence: $\hat{\delta}_{\text{diff}} \geq \mathcal{C}(N_2, x^{**}) - \mathcal{C}(N_1, x^{**})$.

Step 3. Combining:

$$\begin{aligned} \delta_1(N_1) + \hat{\delta}_{\text{diff}} &\geq \delta_1(N_1) + \mathcal{C}(N_2, x^{**}) - \mathcal{C}(N_1, x^{**}) \\ &\geq \mathcal{C}(N_1, x^{**}) + \mathcal{C}(N_2, x^{**}) - \mathcal{C}(N_1, x^{**}) \quad (\text{Step 1: } \delta_1(N_1) \geq \mathcal{C}(N_1, x^{**})) \\ &= \mathcal{C}(N_2, x^{**}) \geq \hat{\delta}_1(N_2) - \epsilon. \end{aligned}$$

451 Since $\epsilon > 0$ is arbitrary, $\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)$.

452 **Corollary 2 (Sandwich bounds).** *If both transfer MIPs are feasible:*

$$\delta_1(N_1) + \delta_{\text{diff}}^* \leq \delta_1(N_2), \quad \delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2).$$

453 *Remark 3 (Complementary regions).* The two transfer quantities explore com-
 454plementary regions: the original δ_{diff}^* searches over N_1 -robust $\cap$ N_2 -foolable in-
 455puts, while $\hat{\delta}_{\text{diff}}$ searches over N_1 -foolable $\cap$ N_2 -foolable inputs. These regions are
 456disjoint. In particular, $\hat{\delta}_{\text{diff}}$ can be negative (if N_1 always has higher confidence
 457than N_2 in $\mathcal{F}_{1 \wedge 2}$), while $\delta_{\text{diff}}^* \geq 0$ by constraint C2.

458 8.5 Numerical Verification

459 We verify Theorem 3 on the counterexample networks from Section 8.2.

460 *Modified transfer value $\hat{\delta}_{\text{diff}}$.* We maximize $x_1 - 4x_2$ subject to $x_1 - x_2 \leq 0.199$
 461 (N_1 -foolable) and $2x_1 - 5x_2 \leq 0.699$ (N_2 -foolable), with $x \in [0, 1]^2$. Setting
 462 $x_2 = 0$: $x_1 \leq \min(0.199, 0.3495) = 0.199$. This yields $\hat{\delta}_{\text{diff}} = \mathbf{0.199}$.

463 *Modified standard-mode value $\hat{\delta}_1(N_2)$.* We maximize $2x_1 - 5x_2$ subject to both
 464 foolability constraints. At $x_2 = 0$: $x_1 \leq 0.199$, so $\hat{\delta}_1(N_2) = 2 \times 0.199 = \mathbf{0.398}$.

Quantity	Value
$\delta_1(N_1)$	0.199
$\hat{\delta}_{\text{diff}}$	0.199
$\delta_1(N_1) + \hat{\delta}_{\text{diff}}$	0.398
$\hat{\delta}_1(N_2)$	0.398
$\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)?$	YES (equality holds)
$\delta_1(N_2)$ (unrestricted)	0.699
Joint gap: $\delta_1(N_2) - \hat{\delta}_1(N_2)$	0.301

Equality holds in Theorem 3 because the binding constraint is simultaneously $\widehat{C1}$ and C3, making x^{**} the argmax for both $\hat{\delta}_1(N_2)$ and $\hat{\delta}_{\text{diff}}$. The joint gap of 0.301 reflects N_2 's foolable region that is N_1 -robust — precisely the regime probed by the original transfer MIP.

8.6 Discussion: Two Complementary Formulations

The original and modified transfer MIPs provide complementary views of the robustness relationship between two networks. Table 3 summarizes their characteristics.

Table 3. Comparison of the two transfer MIP formulations.

Formulation	Feasible region	Bound direction
Original δ_{diff}^*	N_1 -robust $\cap$ N_2 -foolable	$\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*$ (lower bound)
Modified $\hat{\delta}_{\text{diff}}$	N_1 -foolable $\cap$ N_2 -foolable	$\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)$ (upper bound)

The original formulation asks: “By how much can N_2 's confidence exceed N_1 's, at a point where N_1 is provably robust?” Theorem 2 shows this provides a lower bound on $\delta_1(N_2) - \delta_1(N_1)$.

The modified formulation asks: “At points hardest for *both* networks simultaneously, how much more confident is N_2 than N_1 ?” Theorem 3 shows this provides an upper bound on $\hat{\delta}_1(N_2) - \delta_1(N_1)$.

Together, the two quantities bracket the robustness difference from both sides over their respective domains.

9 Tightening Techniques

In this section, we present four complementary techniques for tightening the MIP relaxation. As described in Section 5, the time complexity of the MIP is governed by the number of boolean variables. To enable the MIP solver to reach a solution efficiently, we introduce techniques that reduce the search space by providing tighter bounds and warm-start solutions. The four techniques can be combined freely, progressively trading preprocessing time for faster solver convergence.

9.1 Hyper-Adversarial Attack

Our first technique expedites the tightening of the lower bound by efficiently computing a suboptimal feasible solution and providing it to the MIP solver. Before launching the MIP solver, we run a projected gradient descent (PGD) attack [9] to find a feasible solution (x^*, x'^*) with objective value v^* .

Table 4. Overview of the four tightening techniques.

Technique	Method	Preprocess	Benefit
Hyper-adversarial attack	PGD warm-start	Python PGD	Incumbent cutoff
Perturbation-diff. intervals	Interval propagation	Interval arith.	Activation bounds
Inter-network intervals	Weight-difference bounds	Interval arith.	Cross-network bounds
Dependency analysis	Monotonicity signatures	Symbolic prop.	Ordering constraints

495 This solution is used in two ways. First, as an *incumbent cutoff*: the solver
 496 can prune any branch whose objective value cannot exceed v^* . Formally, we set
 497 the Gurobi incumbent to v^* , enabling the solver to discard large portions of the
 498 search tree.

499 Second, as a *binary warm-start*: for each ReLU binary variable b_j , we compute
 500 the activation sign at the PGD solution and provide it as a hint:

$$b_j^{\text{hint}} = \mathbf{1}\left[z_j^{[l]}(x^*) \geq 0\right].$$

501 These hints guide the solver toward the optimal branching decisions, significantly
 502 reducing the number of explored nodes.

503 9.2 Perturbation-Difference Interval Propagation

504 Our second technique bounds the activation differences between the clean and
 505 perturbed forward passes. For each layer l , we compute an interval $[I^{\text{dn}}[l], I^{\text{up}}[l]]$
 506 bounding the element-wise difference $a^{\langle \text{pert} \rangle}[l] - a^{\langle \text{org} \rangle}[l]$. The same propagation
 507 applies to both the N_1 and N_2 network pairs.

508 *Initialization (input layer).* From Table 1: $I^{\text{dn}}[0] = I^{\text{dn}}$, $I^{\text{up}}[0] = I^{\text{up}}$.

Linear layer propagation. For weight matrix $W^{[l]}$ (the bias cancels in the difference):

$$I^{\text{up}}[l] = (W^{[l]})^+ I^{\text{up}}[l-1] + (W^{[l]})^- I^{\text{dn}}[l-1], \quad (7)$$

$$I^{\text{dn}}[l] = (W^{[l]})^+ I^{\text{dn}}[l-1] + (W^{[l]})^- I^{\text{up}}[l-1], \quad (8)$$

509 where $W^+ = \max(W, 0)$ and $W^- = \min(W, 0)$.

510 *ReLU layer propagation.* We use a linear over-approximation:

$$I^{\text{up}}[l] \leftarrow \max(0, I^{\text{up}}[l-1]), \quad I^{\text{dn}}[l] \leftarrow \min(0, I^{\text{dn}}[l-1]).$$

511 *Added MIP constraints.* At every layer l and neuron j :

$$a^{\langle \text{org} \rangle}_j[l] + I^{\text{dn}}_j[l] \leq a^{\langle \text{pert} \rangle}_j[l] \leq a^{\langle \text{org} \rangle}_j[l] + I^{\text{up}}_j[l]. \quad (9)$$

512 These constraints couple the clean and perturbed forward passes, tightening the
 513 LP relaxation. We next prove their soundness.

514 *Soundness, completeness, and tightness.* We distinguish three properties of
 515 added constraints in a MIP verifier:

- 516 – **Soundness** (no false negatives): a constraint is sound if it is satisfied by
517 every feasible point of the original MIP, so it never removes a real solution.
- 518 – **Completeness** (no false positives): a constraint is complete if it does not
519 introduce spurious feasible points, so the LP relaxation does not admit so-
520 lutions that could not arise from any true (x, x') pair.
- 521 – **Tightness** (quality of approximation): a sound constraint is tight at a point
522 if the bound it provides equals the true value there.

523 The interval constraints (9) are **sound and complete but not always**
524 **tight**. They are sound because the propagated intervals are genuine over-approximations
525 (Theorem 4). They are complete because, being over-approximations, they can
526 only widen the set of allowed activations. They are not always tight because the
527 ReLU propagation step uses a linear over-approximation that is loose when one
528 of the two compared activations is negative and the other is positive.

529 **Theorem 4 (Soundness of perturbation-difference interval propaga-**
530 **tion).** *Let N be a ReLU network with weights $W^{[l]}$, biases $b^{[l]}$. Fix any x and x'*
531 *such that $x'_i - x_i \in [I_i^{\text{dn}[0]}, I_i^{\text{up}[0]}]$ elementwise. Then for every layer l and every*
532 *neuron j :*

$$a_j^{[l]}(x') - a_j^{[l]}(x) \in [I_j^{\text{dn}[l]}, I_j^{\text{up}[l]}],$$

533 where $[I_j^{\text{dn}[l]}, I_j^{\text{up}[l]}]$ are computed by the recursion in (7)–(8) and the ReLU prop-
534 agation step.

535 *Proof.* By induction on l .

536 **Base case** ($l = 0$): $a^{[0]}(x) = x$ and $a^{[0]}(x') = x'$, so $a_i^{[0]}(x') - a_i^{[0]}(x) =$
537 $x'_i - x_i \in [I_i^{\text{dn}[0]}, I_i^{\text{up}[0]}]$ by assumption.

Inductive step — linear layer. Assume $\Delta_i := a_i^{[l-1]}(x') - a_i^{[l-1]}(x) \in$
538 $[I_i^{\text{dn}[l-1]}, I_i^{\text{up}[l-1]}]$ for all i . The pre-activation difference at neuron j is $z_j^{[l]}(x') -$
539 $z_j^{[l]}(x) = \sum_i W_{ji}^{[l]} \Delta_i$. Decompose $W_{ji}^{[l]} = W_{ji}^+ + W_{ji}^-$ with $W_{ji}^+ \geq 0$, $W_{ji}^- \leq 0$.
Then:

$$\begin{aligned} z_j^{[l]}(x') - z_j^{[l]}(x) &\leq \sum_i W_{ji}^+ I_i^{\text{up}[l-1]} + \sum_i W_{ji}^- I_i^{\text{dn}[l-1]} = I_j^{\text{up}[l]}, \\ z_j^{[l]}(x') - z_j^{[l]}(x) &\geq \sum_i W_{ji}^+ I_i^{\text{dn}[l-1]} + \sum_i W_{ji}^- I_i^{\text{up}[l-1]} = I_j^{\text{dn}[l]}, \end{aligned}$$

538 using monotonicity of multiplication by a non-negative (resp. non-positive) scalar.
539 The bias cancels in the difference.

540 **Inductive step — ReLU layer.** Write $\Delta_j^z := z_j^{[l]}(x') - z_j^{[l]}(x) \in [I_j^{\text{dn}[l]}, I_j^{\text{up}[l]}]$.
541 For any $u, v \in \mathbb{R}$: $\max(0, u) - \max(0, v) \leq \max(0, u - v)$ and $\max(0, u) -$
542 $\max(0, v) \geq \min(0, u - v)$. Applying with $u = z_j(x')$, $v = z_j(x)$:

$$\min(0, \Delta_j^z) \leq a_j^{[l]}(x') - a_j^{[l]}(x) \leq \max(0, \Delta_j^z).$$

543 Since $\Delta_j^z \leq I_j^{\text{up}[l]}$, we have $\max(0, \Delta_j^z) \leq \max(0, I_j^{\text{up}[l]})$, and since $\Delta_j^z \geq I_j^{\text{dn}[l]}$, we
544 have $\min(0, \Delta_j^z) \geq \min(0, I_j^{\text{dn}[l]})$. The updated interval $[\min(0, I_j^{\text{dn}[l]}), \max(0, I_j^{\text{up}[l]})]$
545 is exactly what the ReLU propagation step produces.

Corollary 3 (Constraint validity). *Every feasible point $(x, x', \{a^{(m)[l]}\})$ of the MIP already satisfies constraints (9). Adding them therefore does not cut off any feasible solution, but shrinks the LP relaxation polytope, strengthening the dual bound and reducing branch-and-bound effort.*

9.3 Inter-Network Interval Bounds

Our third technique exploits the structural similarity between N_1 and N_2 . When N_2 is obtained by fine-tuning or distilling N_1 , their weights are close, and we can bound the inter-network activation differences.

We compute interval bounds on the activation difference $a^{N_2[l]} - a^{N_1[l]}$ at each layer l , using the weight-difference matrices:

$$\Delta W^{[l]} = W_2^{[l]} - W_1^{[l]}, \quad \Delta b^{[l]} = b_2^{[l]} - b_1^{[l]}.$$

Let $[\alpha^{[l]}, \beta^{[l]}]$ bound $a^{N_2[l]} - a^{N_1[l]}$, and let $\mathbf{u}^{[l-1]}$ be the upper bound on $a^{N_1[l-1]}$ from standard interval arithmetic. The propagation recursion is:

$$\beta^{[l]} = (W_1^{[l]})^+ \beta^{[l-1]} + (W_1^{[l]})^- \alpha^{[l-1]} + |\Delta W^{[l]}| \mathbf{u}^{[l-1]} + \Delta b^{[l]}, \quad (10)$$

$$\alpha^{[l]} = (W_1^{[l]})^+ \alpha^{[l-1]} + (W_1^{[l]})^- \beta^{[l-1]} - |\Delta W^{[l]}| \mathbf{u}^{[l-1]} + \Delta b^{[l]}. \quad (11)$$

The resulting MIP constraints are:

$$a^{N_1[j]} + \alpha_j^{[l]} \leq a^{N_2[j]} \leq a^{N_1[j]} + \beta_j^{[l]}. \quad (12)$$

These constraints are particularly effective when N_1 and N_2 are structurally similar, as the weight differences are small and the resulting intervals are tight.

9.4 Dependency Analysis

Our fourth technique identifies *monotone relationships* between the activations of the clean and perturbed forward passes. We track the direction in which each activation responds to the perturbation using a *dependency signature* $\varphi_j \in \{-1, 0, +1, \text{NaN}\}$: +1 means the activation is monotonically increasing with the perturbation, -1 means decreasing, 0 means unchanged, and NaN means the direction is ambiguous.

Initialization (input layer). The signature depends on the perturbation type:

- ℓ_∞ : $\varphi_i^{[0]} = \text{NaN}$ for all i (each pixel can increase or decrease).
- Brightness: $\varphi_i^{[0]} = +1$ for all i (pixels can only increase).
- Contrast: $\varphi_i^{[0]} = +1$ for all i .
- Occlusion: $\varphi_i^{[0]} = -1$ for $i \in P$, else 0 (occluded pixels decrease, others unchanged).
- Patch: $\varphi_i^{[0]} = \text{NaN}$ for $i \in P$, else 0.

573 *Propagation through a linear layer.* Each output neuron j merges the signa-
 574 tures of its inputs, weighted by the sign of the connecting weight:

$$\varphi_j^{[l]} = \bigoplus_{i: W_{ji}^{[l]} \neq 0} \text{sign}(W_{ji}^{[l]}) \cdot \varphi_i^{[l-1]},$$

575 where $\oplus$ returns the common value if all operands agree, otherwise NaN.

576 *Propagation through ReLU.* If the neuron is always active ($z_j \geq 0$), the sig-
 577 nature is inherited. If the neuron is always inactive ($z_j \leq 0$), the signature is 0.
 578 Otherwise, the signature is NaN.

Added MIP constraints. For each neuron where the signature is determined:

$$\varphi_j^{[l]} = +1 \Rightarrow a^{\langle \text{pert} \rangle}_j^{[l]} \geq a^{\langle \text{org} \rangle}_j^{[l]}, \quad (13)$$

$$\varphi_j^{[l]} = -1 \Rightarrow a^{\langle \text{pert} \rangle}_j^{[l]} \leq a^{\langle \text{org} \rangle}_j^{[l]}, \quad (14)$$

$$\varphi_j^{[l]} = 0 \Rightarrow a^{\langle \text{pert} \rangle}_j^{[l]} = a^{\langle \text{org} \rangle}_j^{[l]}. \quad (15)$$

579 These ordering constraints prune the search space by eliminating variable
 580 assignments that violate the proven monotone relationships.

581 10 Complete MIP Formulation

582 In this section, we present the complete transfer-mode MIP with all four tighten-
 583 ing techniques enabled. The formulation brings together the transfer constraints
 584 from Section 6 and the tightening constraints from Section 9 into a single opti-
 585 mization problem.

586 The objective is:

$$\max_{x, x', \{a^{\langle m \rangle}_j^{[l]}\}, \{b^{\langle m \rangle}_j^{[l]}\}} \mathcal{C}(N_2, x, c) - \mathcal{C}(N_1, x, c) \quad (16)$$

subject to the following constraints:

$$\text{(Perturbation)} \quad x' \in \mathcal{I}_\varepsilon(x) \quad (17)$$

$$\text{(T1)} \quad \mathcal{C}(N_1, x, c) \geq \delta_1 + 10^{-3} \quad (18)$$

$$\text{(T2)} \quad \mathcal{C}(N_2, x, c) \geq \mathcal{C}(N_1, x, c) \quad (19)$$

$$\text{(T3)} \quad \text{Targeted or untargeted attack (Section 7)} \quad (20)$$

$$\text{(ReLU encoding)} \quad \text{Big-}M \text{ for each network pass: } N_1(x), N_2(x), N_2(x') \quad (21)$$

$$\text{(Pert. diff. intervals)} \quad a^{\langle \text{org} \rangle}_j^{[l]} + I^{\text{dn}}_j^{[l]} \leq a^{\langle \text{pert} \rangle}_j^{[l]} \leq a^{\langle \text{org} \rangle}_j^{[l]} + I^{\text{up}}_j^{[l]} \quad (22)$$

$$\text{(Inter-network intervals)} \quad a^{N_1}_j^{[l]} + \alpha_j^{[l]} \leq a^{N_2}_j^{[l]} \leq a^{N_1}_j^{[l]} + \beta_j^{[l]} \quad (23)$$

$$\text{(Dependency constraints)} \quad a^{\langle \text{pert} \rangle}_j^{[l]} \geq a^{\langle \text{org} \rangle}_j^{[l]} \text{ per sign of } \varphi_j^{[l]} \quad (24)$$

$$\text{(Warm-start)} \quad b_j^{\text{hint}} \leftarrow \text{PGD activation signs} \quad (25)$$

587 We note the following about the formulation:

- 588 – The MIP encodes three forward passes: $N_1(x)$, $N_2(x)$, and $N_2(x')$. The pass
 589 $N_1(x')$ is not needed since no constraint involves N_1 's output on the per-
 590 turbed input.
- 591 – Constraints T1–T3 are encoded via the big- M confidence-margin lineariza-
 592 tion described in Section 5.
- 593 – The perturbation-difference interval constraints (6th row) are added by the
 594 second tightening technique (Section 9.2).
- 595 – The inter-network interval constraints (7th row) are added by the third tight-
 596 ening technique (Section 9.3).
- 597 – The dependency constraints (8th row) are added by the fourth tightening
 598 technique (Section 9.4).
- 599 – The binary warm-start (9th row) is provided by the hyper-adversarial attack
 600 (Section 9.1).

601 11 Inference-Time Decision Procedure

602 In this section, we describe how the precomputed transfer guarantee is applied
 603 at inference time. Given the standard-mode value δ_1 (from Section 5) and the
 604 transfer value δ_{diff} (from Section 6), the runtime procedure certifies individual
 605 inputs using only two forward passes.

606 For each input x , the procedure applies a two-stage test:

- 607 1. **Standard-mode check.** Compute $c_1 := \mathcal{C}(N_1, x, c)$. If $c_1 < \delta_1$, the input
 608 cannot be certified as robust for N_1 (or, consequently, for N_2).
- 609 2. **Transfer check.** Compute $c_2 := \mathcal{C}(N_2, x, c)$. If $c_2 - c_1 > \delta_{\text{diff}}$, certify x as
 610 $(\varepsilon, c \rightarrow t)$ -robust for N_2 .

611 By Corollary 1, an equivalent single-threshold condition is:

$$c_1 \geq \delta_1 \text{ and } c_2 > \delta_1 + \delta_{\text{diff}} \implies x \text{ is } (\varepsilon, c \rightarrow t)\text{-robust for } N_2.$$

612 This procedure requires only two forward passes at inference time and no ad-
 613 ditional optimization. The MIP is solved once offline, and the resulting thresholds
 614 δ_1 and δ_{diff} are used for all subsequent inputs.

615 Note that, in the notation of Section 2.5, $\delta_1 = \delta_1(N_1)$ and $\delta_{\text{diff}} = \delta_{\text{diff}}^*$. By
 616 Theorem 2, $\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*$, so the threshold $\delta_1 + \delta_{\text{diff}}$ is always a valid
 617 (conservative) certificate for N_2 . The certificate is strict: there may exist inputs
 618 with $\mathcal{C}(N_2, x, c) \in [\delta_1(N_1) + \delta_{\text{diff}}^*, \delta_1(N_2)]$ that are also robust but not certified
 619 by this procedure. This gap is precisely characterized in Section 8.3.

620 12 Conclusion

621 In this work, we addressed the problem of verifying the transfer of robustness
 622 guarantees between neural networks. We extended VHAGaR with a transfer
 623 mode that encodes the problem as a mixed-integer program, simultaneously
 624 modeling three forward passes — $N_1(x)$, $N_2(x)$, and $N_2(x')$ — to capture the

interaction between two networks under adversarial perturbation. Our approach supports the L_∞ perturbation and six semantic feature perturbations, and employs four complementary tightening techniques to make the MIP tractable.

We established three formal results about the transfer MIP:

1. **Theorem 2:** the original transfer MIP provides a sound lower bound, $\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*$. This enables certifying N_2 's robustness by leveraging N_1 's pre-computed guarantee.
2. **Counterexample** (Section 8.2): the reverse inequality $\delta_1(N_1) + \delta_{\text{diff}}^* \geq \delta_1(N_2)$ is false in general. The gap equals the surplus N_1 -confidence at the transfer optimum ($\mathcal{C}(N_1, x^*) - \delta_1(N_1)$), which is always positive.
3. **Theorem 3:** the modified transfer MIP provides a matching upper bound, $\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)$, over the joint-foolable region $\mathcal{F}_{1 \wedge 2} = \mathcal{F}_1 \cap \mathcal{F}_2$.

Together, the two formulations provide a two-sided characterization of the robustness relationship between any pair of networks: the original MIP establishes how much robustness is guaranteed to transfer, while the modified MIP bounds how much can be gained in the shared vulnerability region.

12.1 Summary of Tightening Techniques

Table 5 summarizes the four tightening techniques and their effects on the MIP.

Table 5. Summary of the four tightening techniques.

Technique	Core method	Effect on MIP
Hyper-adversarial attack	PGD attack $\rightarrow v^*$	Solver incumbent; binary hints b_j
Pert.-diff. interval propagation	$[I^{\text{dn}[l]}, I^{\text{up}[l]}]$ propagation	Bounds $a^{\langle \text{pert} \rangle [l]} - a^{\langle \text{org} \rangle [l]}$ per layer
Inter-network intervals	$[\alpha^{[l]}, \beta^{[l]}]$ via ΔW	Bounds $a^{N_2[l]} - a^{N_1[l]}$ per layer
Dependency analysis	$\varphi^{[l]} \in \{-1, 0, +1, \text{NaN}\}$	Monotone equality/inequality constraints

12.2 Future Work

Several natural extensions of this work remain. First, generalizing the transfer formulation from two classes to the full K -class setting would broaden its applicability. Second, exploring non- ℓ_∞ perturbation sets beyond those already supported could address a wider range of adversarial threat models. Third, developing analogous bounds for backbone-sharing architectures, where N_1 and N_2 share a common encoder, could exploit additional structural constraints. Finally, closing the gap between $\delta_1(N_1) + \delta_{\text{diff}}^*$ and $\delta_1(N_2)$ by searching over the full $\mathcal{F}_2$ region with additional structural constraints is an important direction for obtaining tighter transfer guarantees.

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